CPST 342 Introduction to Web Development

INSTRUCTOR INFORMATION

Instructor Name: Christopher Fulton
Instructor Email: cfulton@luc.edu
Online Office Hours: Appointments are available upon request to offer more flexibility. Please email me if you'd like to schedule a one-on-one session. The last 30 minutes of each class will be dedicated to addressing questions, ideas, and concerns.

COURSE INFORMATION

Course Start Date: 10/17/2026
Course End Date: 12/12/2026
Course Meeting Times: Thursday 6:00 pm – 8:30 pm
Course Location: This is an 8-week, fully online synchronous course via Zoom. The Zoom link can be found in Brightspace.
Course Website: Brightspace is the learning management system where we will host all course materials. All lecture slides, assignment write-ups, course content, and learning materials will be posted in Brightspace. The link below will take you to this semester's course site.

https://brightspace.luc.edu/d2l/home/18427

Course Description and Learning Outcomes:
This course studies the concepts, methods, and tools used in the analysis, design, implementation, testing, and deployment of typical multi-tier web applications. Specific topics include presentations, business, and data access layers, as well as integration with database systems. An introduction to webpage development using JavaScript, jQuery and associated client-side tools.

Outcomes: Experience with commonly used web application development frameworks; Familiarity with database integration and deployment environments; Create webpages using JavaScript and related tools and protocols, and interface a webpage with a database.

Course Objectives:

Upon successful completion of this course, students will be able to:

I. Use development tools used to build applications such as Integrated Development Environments (IDEs) and text editors

II. Utilize a version control tool to manage and deploy source code

III. Use web application development frameworks

IV. Explore database integration and deployment environments

V. Create webpages using JavaScript and related tools and protocols.

VI. Construct an interface that displays information from a database.

VII. Integrate testing tools to ensure code quality.

PRE/COREQUISITE COURSE(S)

Prerequisite: COMP 170 with C- or better. Restricted to students in the School of Continuing and Professional Studies.

OPTIONAL TEXTBOOK

For this course, optional reading will be assigned each week, which you can find in the weekly modules in Brightspace. You can access the book through our school's library website. Listed below are details regarding the book and a direct link to access the eBook once authenticated using your Loyola credentials.

- Book Title: Full Stack Web Development, Philip Ackermann, Rheinwerk Publishing
- URL to directly access the book: https://learning.oreilly.com/library/view/full-stack-web/9781836644934

LEARNER SUPPLIES

Listed below are the items you will need for our course:

- **GitHub Account** - Your GitHub Account will host all your assignments and labs for the course. During the first few weeks of our synchronous class sessions, we will walk through how to create a GitHub account and publish your work
  .
- **Internet Access and Google Chrome browser** - Access to high-speed internet is critical and essential for our weekly online synchronous sessions. In addition to internet access, you will need to have the Google Chrome browser installed. You can find more information about the browser installation at the link below
  - https://www.google.com/chrome/downloads/
- **Reliable Computer and Webcam** - Due to our course being online and technical, it is expected that your computer will have the specifications to handle installing software and the ability to utilize a webcam.
  https://www.luc.edu/its/services/desktoplaptopmobiledevices/technologypurchases/personaltechnologypurchases/personalpurchaserecommendations/

GRADING SCALE

Letter Grade/Percentage Equivalents for SCPS:

A 100% - 95%	B 86% - 88%	C 77% - 79%	D 65% - 70%
A- 92% - 94%	B- 83% - 85%	C- 74% - 76%	F <65%
B+ 89% - 91%	C+ 80% - 82%	D+ 71% - 73%	

GRADING RATIONALE

- Assignments = 30% of final grade | 3 Assignments (30 points each)
- Quizzes/Participation = 20% | 6 Quizzes (10 points each)
- Assigned Reading via FreeCodeCamp.org = 13.33% | 4 Assigned Readings (10 points each)
- Lab Activities = 20% | 4 Lab Activities (15 total points)

- Exams = 16.66% | 1 Final Exam (50 Points)

Total Possible Points: 300

FORMATIVE ASSESSMENT
Assignments (1 - 3), Labs, Assigned Readings

SUMMATIVE ASSESSMENT
Quizzes, Final Exam

SCHEDULE

Session	Course Outcome(s)	Topics/Activities *All readings are due at 6 pm before the weekly session	Assessment(s)
Week 1 Sync. (10/22)	CO I	Introductions and Syllabus Review Understanding Web Architecture and Web Tools Installing development environments	Lab 1 Released
Week 2 Sync. (10/29)	CO I, II	Version Control with Git and GitHub	Assignment 1 Released **Lab 1: Version Control Due (11/1)**
Week 3 Sync. (11/5)	CO III, V	Overview of HTML/CSS/JavaScript	Lab 2 Released **Assignment 1 Due (11/8)**
Week 4 Sync. (11/12)	CO III, V	Introduction to JavaScript ES6, Node.js and NPM **FreeCodeCamp.org Part I (3 Sections)** - **Introduction to Node.js** - **Node.js Core Modules** - **Node Package Manager**	Assignment 2 Released **Lab 2: Page Layout Due (11/15)** **FreeCodeCamp.org Reading Due (11/15)**
Week 5 Sync. (11/19)	CO III	Back-end Frameworks: Routing and Middleware with Express **FreeCodeCamp.org Part II (3 Sections)** - **HTTP and the Web Standards Model** - **Introduction to Express** - **Express Middleware**	Lab 3 Released Assignment 3 Released **FreeCodeCamp.org Reading Due (11/22)**
Week 6 Async.	CO	**[ASYNCHRONOUS – PROJECT WORK TIME]** Interpreting JSON and Understanding REST API **FreeCodeCamp.org Part III (2 Sections)**	**Assignment 2 Due (11/29)** **Lab 3 Due (11/29)** **FreeCodeCamp.org Reading Due (11/29)**

		- **REST API and Web Services** - **Error Handling in Express**	
Week 7 Async.	CO III, IV	**[ASYNCHRONOUS – PROJECT WORK TIME]** Back-end Frameworks: Database Integration, API, JSON **Project Check-ins [15 mins.]**	Lab 4 Released
Week 8 Sync. (12/10)	CO II, VII	Review of App Deployment: Hosting, CI/CD overview.	**Lab 4 Due (12/10)** **Assignment 3 Due (12/10)** *No extension for Lab 4 or Assignment 3 – Must be submitted by Dec. 10 prior to class.* **Final Exam** **Final Exam completed during the synchronous class session*

ATTENDANCE

Attendance and active participation are required for all synchronous class sessions. All required sessions are listed below.

- October 22, 2026 6 pm – 8:30 pm
- October 29, 2026 6 pm – 8:30 pm
- November 5, 2026 6 pm – 8:30 pm
- November 12, 2026 6 pm – 8:30 pm
- November 19, 2026 6 pm – 8:30 pm
- December 10, 2026 6 pm – 8:30 pm

SUCCEEDING IN THIS CLASS

As the instructor, my job in this class is to organize the material coherently, give helpful lectures, provide a framework that combines enough challenge and support for success, and grade reasonably. A reasonable academic expectation is that you spend 2 hours outside of class for every hour spent in class. For a few of your assignments, you may go over this time estimate.

- Complete the assigned reading and review material before each class session.
- Take notes and participate in all the learning activities found in the slide presentations
- Complete labs and assignments to the best of your ability

LATE WORK POLICY

All assignments and labs must be submitted by the due date and time listed on the program. If the program is not turned in by the due date and time listed, the program will not receive points and will be assessed a zero.

Students can receive up to 2 consecutive days beyond the original due date if they notify the instructor that an extension is needed before the due date listed on the program. Once the extension expires, the program will automatically receive a zero if not submitted within 2 consecutive days beyond the original due date. Program extensions will not be permitted if the instructor is notified after the due date. All quizzes and exams must be

submitted by the due date and time.

No make-up or exceptions will be made for quizzes and exams unless explicitly stated otherwise.

ACADEMIC INTEGRITY + PLAGIARISM

Cheating, plagiarism, and any other forms of academic dishonesty will not be tolerated.
This course will adhere to the institution's standards. The link below outlines each standard and sanctions for academic misconduct. Once you navigate to the webpage below, click on the “Academic Integrity” tab to review standards and sanctions.

https://catalog.luc.edu/academic-standards-regulations/undergraduate/

ACCOMMODATIONS

Loyola University Chicago provides reasonable accommodations for students with disabilities. Any student requesting accommodations related to a disability or other condition is required to register with the Student Accessibility Center (SAC). Professors will receive an accommodation notification from SAC, preferably within the first two weeks of class.

Students are encouraged to meet with their professor individually in order to discuss their accommodations. All information will remain confidential.

Please note that in this class, software may be used to audio record class lectures in order to provide equal access to students with disabilities. Students approved for this accommodation use recordings for their personal study only and recordings may not be shared with other people or used in any way against the faculty member, other lecturers, or students whose classroom comments are recorded as part of the class activity. Recordings are deleted at the end of the semester.

For more information about registering with SAC or questions about accommodations, please contact SAC at 773-508-3700 or SAC@luc.edu.

LEARNER SUPPORT

Tech Support HelpDesk: helpdesk@luc.edu 773-508-4487
Student Accessibility Center: https://www.luc.edu/sac/

DISCLAIMER

This syllabus is subject to change at the instructor's discretion with prior student notification.

SCPS Syllabus Addendum

Simple Syllabus Due Dates

Last Updated:

October 5th, 2026

- *Released Syllabus on Oct. 5th, 2026*

October 27th, 2026

- *Updated Due dates for FreeCodeCamp.org. Week 5,6, and 7 now have due dates and sections to complete*
- *Added Week 7 “Project Check-ins 15-mins”*